118990: silicified laminated shale, Referendum Bore

Location and sampling

GLENGARRY (SG 50–12) AMG Zone 50, 696480E, 7088170N

Sampled on 30 May 1996

The sample was taken from the southern side of a small creek bed about 2.5 km east-northeast of Referendum Bore.

Tectonic unit/relations

The sample is of a fine-grained, banded dark grey silicified horizon that has associated thin white friable layers. The shale is interbedded with basaltic lavas near the stratigraphic base of the Killara Formation of the Yerrida Group, and may in part have had a volcaniclastic origin.

Petrographic description

This sample is a dark grey, finely laminated, chert-like rock that is cross-cut by finely banded and crustiform, 2 to 4 mm wide veins that parallel the sedimentary lamination. In thin section, laminae are defined by small variations in grain size and by discontinuous concentrations of 0.05–0.1 mm angular grains of detrital quartz and feldspar (3–5 vol.%). Ovoid, framboid-like aggregates of a finer grained, translucent mineral, up to 0.1 mm in diameter (5–7 vol.%), are probably Ti-oxides. Matrix minerals cannot be resolved under the microscope because the average grain size of the rock is less than 2.5 mm. Veins consist chiefly of a decussate intergrowth of subhedral albite laths, lesser quartz and minor green biotite that average about 0.5 mm in size. Crude crustiform banding is defined by subtle variations in the grain size of vein minerals. Vein margins with the host rock are gradational and marked by a progressive increase in grain size into the vein.

Zircon morphology

Most of the zircons extracted from this sample are generally well-rounded and the surfaces of the grains extensively pitted, consistent with a detrital origin. A wide range of sizes, from less than 75 × 50 μm to more than 350 × 100 μm, and morphologies, from clear and structureless to strongly

zoned grains, are represented. Grain 1 is a 50 × 60 μm, irregular, light brown-yellow, structureless fragment whereas grain 13 is a 60 × 70 μm, structureless yellow intact crystal.

Analytical details

This sample was analysed on 12 October 1996. The counter deadtime was 32 ns. Eight analyses of the CZ3 standard indicated a Pb^*/U calibration error of 1.87 (1σ%). Common-Pb corrections were made assuming Broken Hill common-Pb isotopic compositions for all unknown analyses with the exception of analyses 12.1 and 14.1, for which isotopic compositions determined using the method of Cumming and Richards (1975) were assumed.

Results

Fifteen analyses were obtained from 15 zircons. Results are given in Table 28 and shown on a concordia plot in Figure 30.

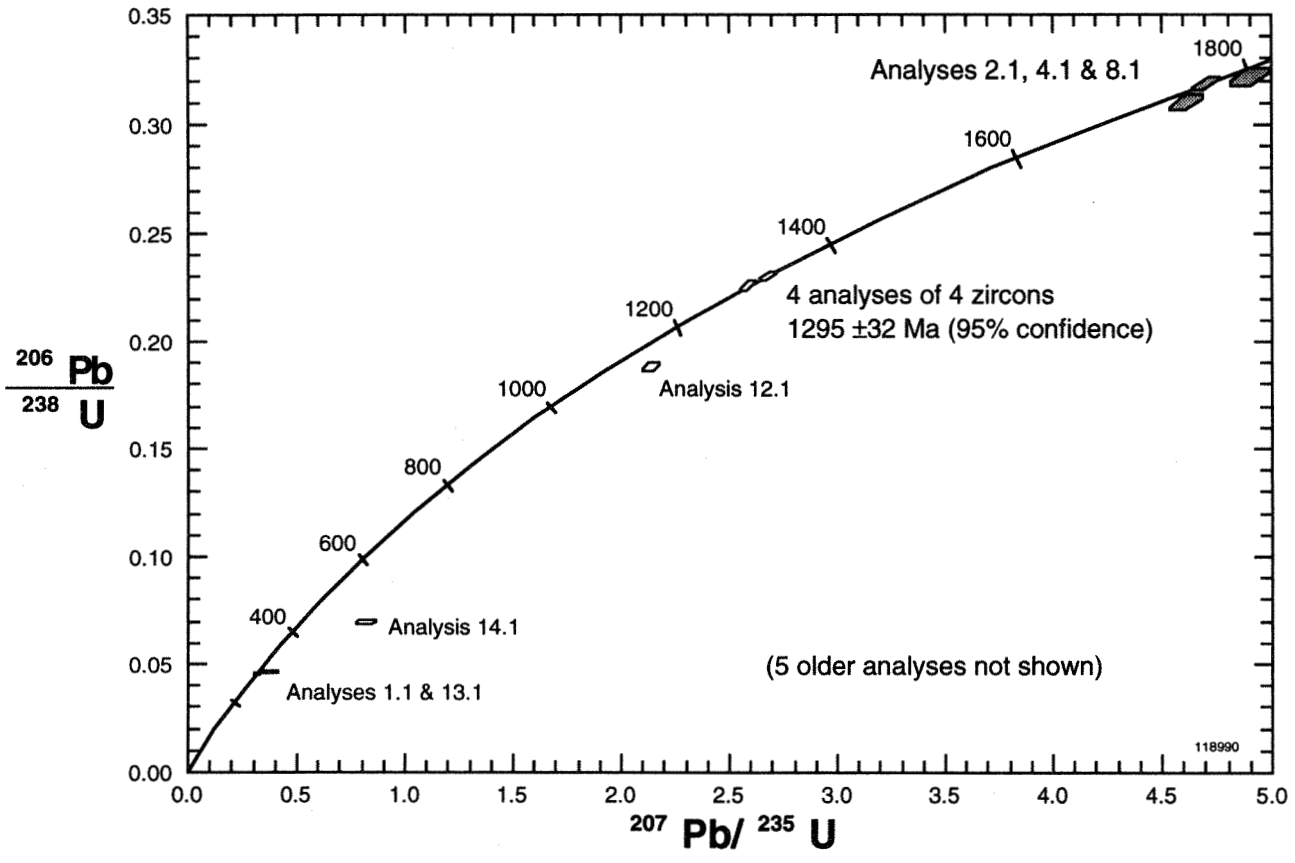

Figure 30. Concordia plot for sample 118990: silicified laminated shale, Referendum Bore

Table 28. Ion microprobe analytical results for sample 118990: silicified laminated shale, Referendum Bore

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	± 1σ	$^{206}Pb/^{238}U$	± 1σ	$^{207}Pb/^{235}U$	± 1σ	*% concordance*	$^{206}Pb/^{238}U$ *Age*	± 1σ	$^{207}Pb/^{206}Pb$ *Age*	± 1σ
1.1	82	61	5	2.298	0.05796	0.00750	0.0460	0.0006	0.367	0.048	55	290	4	528	286
2.1	250	99	83	0.301	0.10789	0.00078	0.3161	0.0033	4.702	0.063	100	1 770	16	1 764	13
3.1	177	81	117	0.073	0.21870	0.00084	0.5732	0.0062	17.284	0.205	98	2 921	25	2 971	6
4.1	120	93	43	0.348	0.10880	0.00118	0.3074	0.0034	4.611	0.075	97	1 728	17	1 779	20
5.1	228	564	227	0.158	0.26300	0.00078	0.6318	0.0067	22.910	0.259	97	3 157	26	3 265	5
6.1	604	9 431	189	9.214	0.14316	0.00226	0.2049	0.0021	4.045	0.081	53	1 202	11	2 266	27
7.1	176	105	117	0.244	0.21095	0.00088	0.5574	0.0060	16.213	0.194	98	2 856	25	2 913	7
8.1	102	131	42	0.426	0.11148	0.00139	0.3188	0.0036	4.901	0.087	98	1 784	17	1 824	23
9.1	210	166	116	0.755	0.21735	0.00107	0.4699	0.0050	14.081	0.173	84	2 483	22	2 961	8
10.1	348	398	97	0.242	0.08493	0.00067	0.2285	0.0024	2.676	0.037	101	1 327	12	1 314	15
11.1	435	60	95	0.202	0.08354	0.00062	0.2244	0.0023	2.584	0.035	102	1 305	12	1 282	15
12.1	446	703	87	1.613	0.08320	0.00120	0.1865	0.0019	2.139	0.040	87	1 102	10	1 274	28
13.1	371	436	21	0.755	0.05019	0.00213	0.0456	0.0005	0.316	0.014	141	287	3	204	98
14.1	595	2 389	72	12.243	0.08589	0.00442	0.0688	0.0008	0.815	0.044	32	429	5	1 336	100
15.1	216	125	94	0.227	0.13632	0.00082	0.3889	0.0041	7.310	0.093	97	2 118	19	2 181	10

Interpretation

Analyses 1.1 and 13.1 are concordant and have a pooled $^{206}Pb/^{238}U$ age of 283 ± 2 Ma. Four additional analyses (10.1, 11.1, 12.1 and 14.1) displaying varying degrees of discordance, have $^{207}Pb/^{206}Pb$ ratios corresponding to an age of 1295 ± 32 Ma (chi-squared = 1.47). The zircons of both these groups may have been derived from the thin albite–quartz–biotite veins identified in this sample. Alternatively, they may be contaminants incorporated into the sample either in the field or during sample processing. A further three concordant analyses (2.1, 4.1 and 8.1) have $^{207}Pb/^{206}Pb$ ratios corresponding to an age of 1779 ± 70 Ma (chi-squared = 2.04). The high chi-squared value of this population suggests that these zircons may have been derived from multiple igneous sources, or have been metamorphically (or hydrothermally) disturbed. Analysis 2.1 has a $^{207}Pb/^{206}Pb$ age of 1764 ± 13 Ma (1σ error) and provides a maximum age for deposition of the sedimentary precursor to the shale. The remaining analyses indicated ages between c. 2180 and 3262 Ma.

STRATIGRAPHIC REFERENCE:

PIRAJNO, F., ADAMIDES, N. G., and FERDINANDO, D. D., 1998, Geology of the Glengarry 1:100 000 sheet: Western Australia Geological Survey, 1:100 000 Geological Series Explanatory Notes.

Recommended reference for this publication:

NELSON, D. R., 1997, 118990: silicified laminated shale, Referendum Bore; in Compilation of SHRIMP U–Pb zircon geochronology data, 1996: Western Australia Geological Survey, Record 1997/2, p. 122–125.
OR
NELSON, D. R., 1997, 118990: silicified laminated shale, Referendum Bore; Geochronology dataset 435; in Compilation of geochronology data, June 2006 update: Western Australia Geological Survey.

Data obtained: 12/10/1996; Data released: 21/08/1997